AP CALCULUS AB

Unit 1 • Lesson 1-8 • Guided notes

Lesson 1-8 • MVT

Guided notes • Fill in the blanks and work the examples in the boxes

Learning goals

- State the Mean Value Theorem and check its hypotheses.
- Find the value(s) of c guaranteed by the MVT.
- Apply the MVT to tables and contexts.

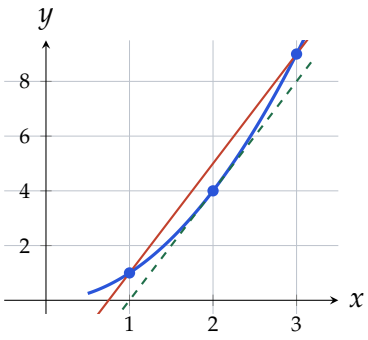
Vocabulary. Mean Value Theorem (MVT), differentiable, open interval, secant slope, tangent slope.

Mean Value Theorem

If f is _____ on $[a,b]$ and _____ on (a,b), then there is at least one c in (a,b) such that

$$f'(c) = \frac{\qquad\qquad}{\qquad\qquad}.$$

Some tangent line is _____ to the secant line: at some instant, the instantaneous rate of change equals the average rate of change.



$f(x) = x^2$ on $[1,3]$: secant slope 4; the tangent at $c = 2$ has slope 4 too.

Example 1 • Finding c

Find the value(s) of c guaranteed by the MVT for (a) $f(x) = x^2$ on $[1,3]$; (b) $g(x) = x^3$ on $[0,3]$; (c) $k(x) = \frac{1}{x}$ on $[1,4]$. (Use $f'(x) = 2x$, $g'(x) = 3x^2$, $k'(x) = -\frac{1}{x^2}$ from Lesson 1-7.)

Example 2 • When the MVT does not apply

Explain why the MVT does not apply: (a) $f(x) = |x|$ on $[-1, 1]$; (b) $g(x) = \frac{1}{x}$ on $[-1, 1]$.

Example 3 • A table in context

A car's position $s(t)$ (miles) is differentiable, with $s(0) = 0$ and $s(2) = 130$ (t in hours). Must the car's speed have been exactly 65 mph at some time?

Common mistakes

- Forgetting to check differentiability (corners, cusps) or continuity.
- Giving a c that is not inside the open interval.

Check your understanding

1. For $f(x) = x^2$ on $[0, 4]$, the MVT gives $c =$ ____.
2. If $f(2) = f(5)$ and f satisfies the hypotheses, then $f'(c) =$ ____ for some c in $(2, 5)$.